The eventual value of h_4, and improved bounds for the Choi–Erdős–Szemerédi pairwise-sums problem (Erdős #866)

Draft v5 — release freeze, 2026-07-08 (C5P2/C6P2 gate execution). Supersedes Draft v4 (same path): no mathematical change — quotes the completed canonical verification counts (G1: 298/298 cells, 0 failures; §9.1 and Appendix B), records the multi-run assembly provenance of that pass honestly, and adds the fresh 2026-07-08 openness kill-check (§13). Release gate G1–G3 GREEN on disk (`check_gate.py`). Cells landed after the 298-cell freeze (h4 65–68, g5 24–25) remain **outside** the claimed ranges. Draft v4 (C6-P2, 2026-06-13) added §10 — the C5-P1 g-side per-anchor collapse ((†), Theorem Z, the conditional capacity ceiling, and the in-flight n-free enumeration), graded honestly per part, **no new claim**. Draft v3 superseded v2 (C4-P4, 2026-06-12); v2 superseded v1 (`C3P4-hygiene-paper/paper/draft-866.md`); v1 upgraded `P5-hygiene/paper/skeleton-866.md`. C4-campaign upgrades folded in: the exact-value theorem for h_4 is now **kernel-verified at** $N_0 = 331{,}777 = 24^4 + 1$ (`Erdos866.h4_eq_4`, the per-o-pair balanced-window collapse, §6 — both halves of Thm 1.3 now [K]), and the general-k constants of §8 are at paper grade (C4-P3 write-out + audit). Evidence grades: [**K**] kernel-grade (Lean 4, sorry-free, axioms exactly `[propext, Classical.choice, Quot.sound]`, fresh audits `lean/scripts/auditC4P1.log` + `c4p1_fullbuild.log` (the h4_eq_4 port, 8039 jobs green) and `lean/scripts/auditC4Synth.log` + `c4synth_build.log` (C4 synthesis: from-scratch rebuild, `.lake/build` deleted first; 28/28 declarations on the three base axioms or fewer, zero `sorryAx`)); [**P**] paper-grade (written proof + exact-arithmetic machine-verified numerics + independent re-verification); [**E**] exact-computational (SAT certificates, dual-engine UNSAT + DRAT/LRAT archive checked by a formally verified checker); [**D**] derivation-grade (derivation in hand, full write-out + audit pending — not headlined). Every claim states **g vs h** explicitly (PROBLEM.md A2).

Abstract

For $A \subseteq \{1, \dots, 2n\}$, let $g_k(n)$ be the least m such that $|A| \geq n + m$ forces k distinct integers $b_1, \dots, b_k$ (not necessarily in A, at most one non-positive automatically) with all $\binom{k}{2}$ pairwise sums $b_i + b_j \in A$, and let $h_k(n)$ be the variant requiring the b_i to be distinct **positive** integers. Choi, Erdős and Szemerédi (1975) proved $h_4(n) = O(1)$ — "we were too lazy to determine t" (Erdős, 1972) — and the best published bounds were $3 \leq h_4(n) \leq 2270$, the upper bound formally verified in Lean (van Doorn–Aristotle, 2026). We determine the eventual value of this constant under the modern (positive-variant) convention:

$h_4(n) = 4$ **for all** $n \geq 331{,}777$,

and this exact-value theorem is itself **verified by the Lean 4 kernel** (`Erdos866.h4_eq_4`, sorry-free, against van Doorn's public formalization), with $h_4(n) = 4$ also at every computed value $n \in \{3, 4\} \cup [6, 64]$ and $h_4(5) = 5$. En route we prove $4 \leq h_4(n) \leq 1000$ for all $n \geq 3$, both halves **verified by the Lean 4 kernel** against van Doorn's public formalization; the lower bound also yields the first strict separation $g_4(n) < h_4(n)$ beyond $k = 3$, likewise kernel-verified. For five summands we improve the g_5 upper constant by a factor 32, from van Doorn's 113,591,719 to 3,519,219 (g-side), by removing the disjoint-representations loss from the CES recursion — this bound too is now **verified by the Lean 4 kernel** (`gFun 5 n < 3519220`, sorry-free, 34× below the prior Lean constant $1.2 \cdot 10^8$) — and we improve the h_5 lower bound from $\log_2 n$ to $\#\{\text{Fibonacci} \leq n\} + 1$

(asymptotic factor $1/\log_2 \varphi \approx 1.4404$), the family-side inequality kernel-verified. We further compute exact values of g_k, h_k ($k \le 6$) on initial segments — in particular $g_5(n) = 4$ for $15 \le n \le 23$, bearing directly on van Doorn's question of whether $g_5(n) \le 4$ for all large n — each nontrivial cell certified by dual SAT engines and an archived DRAT/LRAT proof checked by the formally verified checker cake_lpr. Toward that question we develop (without claiming the theorem) a g-side structural reduction: a one-window per-anchor dichotomy, an all-low/all-top zone theorem, and a conditional capacity ceiling that shrinks the open middle band to $d \in [5, 19]$ (§10). All results are produced and verified under an exact-arithmetic / proof-certificate doctrine described in §12.

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1. Introduction

1.1 The problem

Erdős problem #866 (erdosproblems.com/866, citing [CES75], [Er72], [Er92c]) asks:

> Let $k \ge 3$ and $g_k(N)$ be minimal such that if $A \subseteq \{1, \dots, 2N\}$ has $|A| \ge N + g_k(N)$ then there exist integers $b_1, \dots, b_k$ such that all $\binom{k}{2}$ pairwise sums are in A (but the b_i themselves need not be in A). Estimate $g_k(N)$.

Throughout we use van Doorn's modern formulation [vD26], identical to the public Lean development [L866]: the b_i are pairwise **distinct** integers (without distinctness the problem trivializes), and at most one b_i can be non-positive (two non-positives sum to $\le 0 \notin A$) — this is a derived fact, not a clause. The **positive variant** $h_k(n)$ requires all $b_i \ge 1$. The two functions genuinely differ: $g_3 = 1 < 2 = h_3$, and $h_5 \asymp \log n$ while $g_5 = O(1)$ [vD26]. Basic structure: $g_k \le h_k \le g_{k+1}$ for $k \ge 3$ [vD26 eq. (2); Lean `gFun_le_hFun_le_gFun_succ`].

History. CES75 proved g_3-type and h_3-type results ($h_3 = 2$ for $n \ge 4$), boundedness of the four-summand constant ($h_4 = O(1)$; "we were too lazy to determine t", [Er72] p. 83), $h_5 \asymp \log n$, $g_6 \asymp h_6 \asymp \sqrt{n}$, and the general bounds $t_{k+1} \le 2^k n^{1-2^{-k}}$ and $h_k(n) > n^{1-\epsilon}$ for large k. Van Doorn [vD26] introduced the g/h split, proved $g_3(n) = 1$ ($n \ge 3$), $g_4(n) = 3$ ($n \ge 2$), $g_5(n) < 1.2 \cdot 10^8$, $h_4(n) \le 2270$, and $g_k \le h_k < 4n^{1-2^{2-k}}$; the entire development [vD26] is formalized sorry-free in Lean 4 [L866] (formal author: Aristotle, Harmonic; 3494 lines, Mathlib only). Caveats inherited from the sources: CES75's lower-bound examples for $k \in \{3, 5\}$ silently assume positive b_i and are wrong for g_k (PROBLEM.md A2; e.g. $b = (-1, 2, 3, 5, 6)$ has all ten pairwise sums in odds $\cup$ powers-of-2); [Er72] demanded the *sums* be distinct, a strictly stronger convention than the modern one (A6). Our exact-value theorem for h_4 is stated under the modern convention ([vD26]/Lean `hFun`), and we say so explicitly wherever it is claimed.

1.2 Results

#	Statement (g/h explicit)	Was	Grade	Where
Thm 1.1	$h_4(n) \le 1000$ for all $n \ge 1$	2270 [L866]	**[K]** `h4_le_1000`	§5

#	Statement (g/h explicit)	Was	Grade	Where
Thm 1.1′	$h_4(n) \le 1529$, from Ruzsa's published weak-Sidon bound only	2270	[P] (companion; not separately ported)	§5 (Thm 5.5)
Thm 1.2	$h_4(n) \ge 4$ for all $n \ge 3$; hence $g_4(n) < h_4(n)$ for $n \ge 3$ — first strict g/h separation beyond $k = 3$	3 (via g_4)	**[K]** `four_le_hFun_four`, `gFun_four_lt_hFun_four`	§3
Thm 1.3	$h_4(n) = 4$ **for all** $n \ge 331{,}777$ (positive variant, modern convention) — the eventual value of the constant CES75 left undetermined	$[3, 2270]$	**[K]** `h4_eq_4` (both halves; corollaries `h4_eq_4_charter`/`h4_eq_4_c2` pin the superseded paper-grade constants 4,593,324 and $8 \cdot 10^7$)	§6
Thm 1.4	$g_5(n) \le 3{,}519{,}219$ for all $n \ge 1$ (g-side)	113,591,719 [vD26]; Lean constant $1.2 \cdot 10^8$	**[K]** `g5upper_star_charter` (`gFun 5 n < 3519220`, all n); Lemma 7.1 induction fully proved in Lean at general k	§7
Thm 1.5	$h_5(n) \ge z(n) + 1$ for all $n \ge 1$, $z(n) := \#\{\text{Fibonacci} \le n\}$; $z(n) \ge \lfloor \log_2 n \rfloor + 1$, asymptotic factor $1/\log_2 \varphi \approx 1.4404$ over the standing bound	$> \log_2 n$ [L866]	**[K]** `fibCnt_lt_hFun_five` (family inequality, all n); the z-vs-$\log_2$ comparison is elementary + exhaustively checked to 10^6, not yet formal	§3

#	Statement (g/h explicit)	Was	Grade	Where
Thm 1.6	Exact values ($k \le 6$ tables): $g_5(n) = 4$ for $15 \le n \le 23$; $h_4(n) = 4$ on $\{3,4\} \cup [6,64]$, $h_4(5) = 5$; $g_4(60) = 3$ with unique extremum = vD's family; $h_5(n) = z(n) + 1$ for $13 \le n \le 26$; uniqueness of the Thm-1.2 extremum for $11 \le n \le 64$	vD26 search: no counterexample to $g_5 \le 5$, $n \le 15$	[E]	§9
Thm 1.7	general $k \ge 4$, explicit $N(k) = k^4 \cdot 2^{4k+27}$: $g_k(n) < 2n^{1-2^{2-k}}$, $h_k(n) < 2.25\, n^{1-2^{2-k}}$ for $n \ge N(k)$	$4n^{1-2^{2-k}}$ [vD26]	[P] — full write-out + audit `lenses/C4P3-allk-writeup/T2-ALLK.md`, machine numerics `verify_allk.py` 272 checks PASS (re-run at C4 synthesis); the recursion core (Lemma 7.1 and its extraction lemma) is proved in Lean at general $k \ge 3$; no Lean object for the all-k statement itself	§8

Two results are, to our knowledge, the first of their kind for this problem: Thm 1.3 is the first exact-value theorem for any unbounded-n regime of h_4 (CES75 Theorem 2 proved existence of the constant; no source computed it) — and it is itself kernel-verified — and Thm 1.2 + Thm 1.1 give the first kernel-verified two-sided interval $h_4(n) \in [4, 1000]$ ($n \ge 3$; upper half for all $n \ge 1$). Every constant this paper headlines as beating a published bound, and the exact-value theorem itself, is Lean-verified: Thms 1.1, 1.2, 1.3, 1.4 and the family inequality of Thm 1.5 are all [K]. Beyond the numbered results, §10 reports a structural program toward $g_5 = 4$ (per-anchor dichotomy (†), Theorem Z, conditional capacity) at explicitly sub-[K] grades and with no exact-value claim — it is included for its mathematical content and as the audited state of the open problem, not as a headline result.

1.3 Formalization status (binding, audited 2026-06-12/13 (C4–C5 sessions))

Our Lean 4 extension project builds against the byte-pinned upstream development [L866] (sha256 `043731e3...9845e`, re-hashed at the C4-P1 and C4-synthesis audits; upstream files never modified). From-scratch rebuild (`.lake/build` deleted first): 8039 jobs green (`lean/scripts/c4synth_build.log`). Kernel-verified, axioms exactly `[propext, Classical.choice, Quot.sound]` (one `decide`-backed certificate drops `Classical.choice`), zero `sorryAx` in any root-imported module (audits `lean/scripts/AuditC4P1.lean` → `auditC4P1.log` and `AuditC4Synth.lean` → `auditC4Synth.log`, 28/28 declarations, with compile-time statement pins incl. $331777 = 24^4 + 1$ and `hFun` = upstream `sInf` definition by `rfl`; independently: the C3-synthesis audit `AuditC3Synth.lean`):

- `Erdos866.h4_eq_4 : ∀ n, 331777 ≤ n → hFun 4 n = 4` (Thm 1.3, both halves; module `Erdos866/H4Dichotomy.lean`, imported by the root);
- `Erdos866.g5upper_star_charter : ∀ n, gFun 5 n < 3519220` (Thm 1.4 at the charter constant) with the pinned target `Erdos866.T1TargetG5_closed_charter`, plus the fallback `g5upper_star : ∀ n, gFun 5 n < 3520600`; the engine is the fully proved Lemma 7.1 (`Erdos866.lemmaA`, general $k \geq 3$) and its general-k extraction lemma `Erdos866.ceslemgeneral_star`;
- `h4_le_1000 : ∀ n, 0 < n → hFun 4 n ≤ 1000` (Thm 1.1) and the pinned target `Erdos866.T1TargetH4_closed_1000`;
- `Erdos866.four_le_hFun_four : ∀ n, 3 ≤ n → 4 ≤ hFun 4 n` (Thm 1.2);
- `Erdos866.gFun_four_lt_hFun_four : ∀ n, 3 ≤ n → gFun 4 n < hFun 4 n`;
- `Erdos866.fibCnt_lt_hFun_five : ∀ n, fibCnt n < hFun 5 n` (Thm 1.5 family inequality);
- `Erdos866.key_ineq_star / key_ineq_star_tuned / key_ineq_star_charter / fStar5_le_poly` (the (xineq*) numeric layer of Thm 1.4).

Honestly flagged as **not formal**: Thm 1.1′ (companion only); the z-vs-$\log_2$ comparison in Thm 1.5; the all-k statement of §8 (paper-grade; its recursion core *is* formal); and §9 (which is certificate-checked computation, §9.1, not Lean). The former principal trust locus — the hand-proved structural reduction of Thm 1.3 — was discharged by the Lean port: `structural_case` is closed sorry-free on the per-o-pair chain (§6), and the module is imported by the sorry-free root.

2. Notation

$n \geq 1$; window $W := \{1, \dots, 2n\}$; $A \subseteq W$ with $|A| = n + s$ (s = **surplus**). $g_k(n), h_k(n)$ per §1.1 (= Lean `gFun`/`hFun`). A *(g,k)-config* is k distinct integers b_i with all pairwise sums in A; an *(h,k)-config* additionally has all $b_i \geq 1$. A_0 := even elements of A. For the central-window proof (§5): $C_4 := 1000$ (resp. 1529), $\tau := |A_0| - C_4 \geq 0$, O_{miss} := odd numbers of W missing from A ($|O_{miss}| \leq \tau$). For the recursion (§7): φ = forbidden-set size, x = diameter, $C_5 := 3{,}519{,}219$. Fibonacci normalization (Thm 1.5): $F_1 = 1$, $F_2 = 2$, $F_j = F_{j-1} + F_{j-2}$ (distinct values $1, 2, 3, 5, 8, \dots$); $z(n) = \#\{j : F_j \leq n\}$. The Thm-1.2 family is A_n^*; lab notation $m_k^v(n)$ = maximum config-free size, $v \in \{g, h\}$, so $v_k(n) = m_k^v(n) + 1 - n$. (This applies the skeleton's collision ledger; lens-internal t, C, M are renamed τ, C_4/C_5, $|O_{miss}|$ throughout.)

3. Lower bounds by parity (h-side; Thms 1.2, 1.5)

Theorem 3.1 (= 1.2, [K]). For $n \geq 3$ the set $A_n^* := \{1, 3, \dots, 2n-1\} \cup \{2,\ 2n-2,\ 2n\}$, of size $n + 3$, contains no (h,4)-config. Hence $h_4(n) \geq 4$.

Proof. The even elements are exactly $E = \{2, 2n-2, 2n\}$. Suppose $b_1 < b_2 < b_3 < b_4$ positive with all six sums in A_n^*; split by parity (p, q), p odd count. $(4,0)/(0,4)$: the five distinct values $b_1{+}b_2 < b_1{+}b_3 < b_2{+}b_3 < b_2{+}b_4 < b_3{+}b_4$ are even, but $|E| = 3$. $(3,1)/(1,3)$: the three same-parity b's give three distinct even sums $= E$ exactly, forcing smallest sum $= 2$; but two distinct positive odds sum to ≥ 4 (evens to ≥ 6). $(2,2)$: odds $o_1 < o_2$, evens $e_1 < e_2$; the even sums $o_1{+}o_2$, $e_1{+}e_2 \geq 4 > 2$ lie in $\{2n-2, 2n\}$, so $(o_1{+}o_2) + (e_1{+}e_2) \geq 4n-4$; the cross sum $o_2 + e_2$ is odd, $\leq 2n-1$, so $o_1{+}e_1 \geq (4n-4) - (2n-1) = 2n-3$ and $(o_2{-}o_1) + (e_2{-}e_1) = (o_2{+}e_2) - (o_1{+}e_1) \leq 2$ — but distinct odds and distinct evens each differ by ≥ 2. ■

Lean: `four_le_hFun_four` (witness card + config-freeness via a single `omega` over the 6 sum-memberships, 6 distinctness, 4 positivity facts). Machine checks: config-freeness of A_n^* for $3 \leq n \leq 400$ (Python DFS) and $n \leq 1000 \cup \{1200, 1500, 2000\}$ (independent Rust checker).

Corollary 3.2 ([K]). $g_4(n) = 3 < 4 \leq h_4(n)$ for $n \geq 3$: the $b_1 \leq 0$ escape is *necessary* at surplus 3. This is the first strict $g_k < h_k$ separation beyond $k = 3$ ($g_3 = 1 < 2 = h_3$ was known [vD26]); contrast $k = 5$, where the separation is asymptotic ($g_5 = O(1)$, $h_5 \asymp \log n$). Lean: `gFun_four_lt_hFun_four` (uses upstream `g4`).

Theorem 3.3 (= 1.5, family half [K]). For all $n \geq 1$ (Lean: all n), $B_n := \{1, 3, \dots, 2n{-}1\} \cup \{2F_j : F_j \leq n\}$, of size $n + z(n)$, contains no (h,5)-config. Hence $h_5(n) \geq z(n) + 1$.

Proof. Domination: for Fibonacci values $f_a < f_b < f_c$ we have $f_a + f_b \leq f_c$ (if $f_b = F_j$ then $f_a \leq F_{j-1}$, $f_c \geq F_{j+1} = F_j + F_{j-1}$). Among 5 positive b_i three share parity, say $x_1 < x_2 < x_3$; their pairwise sums are even elements of B_n, i.e. $2f_a = x_1{+}x_2 < 2f_b = x_1{+}x_3 < 2f_c = x_2{+}x_3$ with $f_a < f_b < f_c$ Fibonacci. But $2f_a + 2f_b - 2f_c = 2x_1 \geq 2$, i.e. $f_a + f_b > f_c$ — contradiction. ■

Lean: `fibCnt_lt_hFun_five`, with `fibCnt n` = number of Fibonacci *values* in $[1, n]$ and engine `fib_triple_bound`; first formal improvement of the h_5 lower constant over upstream `h5lower` ($\log_2 n < h_5(n)$).

Corollary 3.4 ([P]). $F_j \leq 2^{j-1}$ gives $z(n) \geq \lfloor \log_2 n \rfloor + 1$, so $h_5(n) \geq \lfloor \log_2 n \rfloor + 2$, beating the standing bound pointwise by ≥ 1 for every $n \geq 2$ (exhaustively checked $n \leq 10^6$; min margin exactly 1) and asymptotically by $1/\log_2 \varphi = 1.4404\dots$ (since $z(n) = \log_\varphi n + O(1)$). *(The comparison lemma $\lfloor \log_2 n \rfloor + 1 \leq z(n)$ is not yet formalized — C4-P5(a).)*

Proposition 3.5 (extremal class, [P]). Call an even set $E = \{e_1 < \cdots < e_m\} \subseteq [2, 2n]$ *dominated* if $e_{i-2} + e_{i-1} \leq e_i$ ($i \geq 3$). Every $A =$ (all odds) $\cup$ (dominated E) is (h,5)-config-free, and conversely $m \leq z(n)$ for dominated E (induction: $e_i/2 \geq F_i$). So $n + z(n)$ is the exact maximum size of an "all-odds + dominated-evens" certificate; the maximizers match every all-odds extremal set found by exact enumeration for $13 \leq n \leq 16$.

4. The weak-Sidon toolbox

$S \subseteq \mathbb{Z}$ finite is **weak Sidon** if no four pairwise-distinct $a, b, c, d \in S$ have $a + b = c + d$.

Lemma 4.1 (energy; cited verified ingredient [K]). $S \subseteq \{1, \dots, N\}$ weak Sidon, $|S| = s \Rightarrow$ for every integer $\nu \geq 1$: $s^2\nu \leq (N{+}\nu{-}1)(3s{+}\nu{-}1)$. *Provenance:* verbatim Lean `weak_sidon_key_ineq` (Upstream.lean:889); mathematical content in the proof of Ruzsa [Ru93, Thm 4.6] (double-count fibers of $S + [1, \nu]$; Cauchy–Schwarz; weak Sidon forces difference multiplicities ≤ 2 with $\leq s$ doubled differences).

Lemma 4.1R (published fallback). Weak Sidon $S \subseteq [1, N]$ has $|S| \leq \sqrt{N} + 4N^{1/4} + 11$ [Ru93, Thm 4.6; Lean `weak_sidon_bound`].

Lemma 4.2. If S is not weak Sidon there are $a_1 < a_2 < a_3 < a_4$ in S with $a_1 + a_4 = a_2 + a_3$ (the equal-sum pairing must match extremes).

Lemma 4.3 ((Φ,ν)-violation). If $\Phi^2\nu > (N_0 + \nu - 1)(3\Phi + \nu - 1)$ and $|S| \geq \Phi$ with S in a window of length N_0, then S is not weak Sidon, yielding the quadruple of 4.2. Anti-monotone in N_0.

Lemma 4.4 (k=4 chain). Let $E \subseteq 2\mathbb{Z}_{\geq 2}$ with diameter $\leq x$, and suppose (Φ, ν) violates the window $x/2 + 1$ and $r(r-1) > 2x(\Phi - 1)$ where $r = |E|$. Then E contains an (h,4)-config. *Proof sketch:* a popular difference d gives $|E \cap (E - d)| \geq \Phi$ after an alternating-path disjointification; apply 4.3 to the halved set; lift the additive quadruple to $(b_1, \ldots, b_4) = (A_1/2,\ A_1/2{+}A_2{-}A_1,\ A_1/2{+}A_3{-}A_1,\ A_1/2{+}d)$. Lean analogue: `ceslem_k4_chain`; in the port the new lemma `not_weak_sidon_of_phinu` feeds `weak_sidon_key_ineq` directly with the certificate's explicit ν on a halved-and-translated Φ-subset.

5. Theorem 1.1: $h_4(n) \leq 1000$ — asymmetric central window [K]

Statement (positive variant): every $A \subseteq \{1, \ldots, 2n\}$ with $|A| \geq n{+}1000$ admits four distinct positive integers with all six pairwise sums in A. Kernel-verified as `h4_le_1000`. The proof is van Doorn's §7 skeleton with two changes: an **asymmetric central window** (change 1) and the **energy-form weak-Sidon base** 4.1 in place of 4.1R (change 2). Set $C_4 = 1000$, $\tau = |A_0| - C_4 \geq 0$, $|O_{miss}| \leq \tau$.

Lemma 5.1 (central element). If $2m \in A$ with $4\tau + 3 \leq m \leq n - 2\tau - 2$, then A has an (h,4)-config. *Proof.* Candidates $b_3 \in K := \{m - 2 - 2i : 0 \leq i \leq 2\tau\}$ ($|K| = 2\tau + 1$, all $\equiv m \bmod 2$); set $b_4 := 2m - b_3$ and take $(b_3,\ m-1,\ m+1,\ b_4)$. The two even sums are $2m \in A$; the four cross sums are odd and (using $8\tau + 6 \leq 2m \leq 2n - 4\tau - 4$) land in $[1, 2n-1]$. Each missing odd kills ≤ 2 candidates ($y < 2m$ only via s_1, s_2; $y > 2m$ only via s_3, s_4), so $\leq 2\tau < |K|$ are bad. ■ *(The window's lower edge is forced by positivity of b_3; the upper edge only needs $2m + 4\tau + 3 \leq 2n - 1$ — hence the asymmetry, which is what shrinks the high interval in Case 2 below.)*

Theorem 5.2 (assembly). If no even element of A is central (Lemma 5.1 empty), then A_0 splits into low $\subseteq [2,\ 8\tau + 4]$ (diameter $\leq 8\tau + 2$) and high $\subseteq [2n - 4\tau - 2,\ 2n]$ (diameter $\leq 4\tau + 2$). A numeric certificate supplies, for each τ, tuples $(\Phi_L, \nu_L, \psi_L, \Phi_H, \nu_H, \psi_H)$ satisfying the verification conditions V2–V5 (each (Φ, ν) violates its window per 4.3, and ψ-cardinality forces the chain 4.4); pigeonhole gives $|\text{low}| \geq \psi_L$ or $|\text{high}| \geq \psi_H$ whenever $|A_0| \geq \psi_L + \psi_H - 1$, and the certificate guarantees $\max_\tau(\psi_L + \psi_H - 1 - \tau) = 1000$. For $\tau \leq 4 \cdot 10^5$ this is a 157-block tile (`cert_energy.csv`); for $\tau > 4 \cdot 10^5$ the closed-form tail (Lemma 5.3). ■

Lemma 5.3 (tail). With $u = \lceil \sqrt{N_0} \rceil$, the tuple $\Phi = 2u$, $\nu = 4u$, $\psi = \text{isqrt}(4xu) + 2$ satisfies V2–V5 for every $\tau > 4 \cdot 10^5$ (three exact rational inequalities, polynomial positivity checked for $u \in [7, 5000]$ and asymptotically).

Proposition 5.4 (route-optimality, [P]). $C_4 = 1000$ is exactly optimal for this window/lemma decomposition: 999 fails at $\tau = 2648$ (the maximum $\psi_L + \psi_H - 1 - \tau = 1000$ is attained at $\tau \in \{2647, 2648\}$ — a tie, recorded in machine artifacts). Likewise 1529 is exact for the Ruzsa-only variant ($\tau = 3559$).

Theorem 5.5 (= 1.1′, [P]). The same skeleton from Lemma 4.1R alone gives $h_4(n) \leq 1529$ (177-block certificate `cert_ruzsa.csv`). This was the port fallback; the energy form landed, so it remains a paper-grade companion for readers who want only published ingredients.

Verification. (i) Lean kernel: `h4_le_1000` sorry-free, axioms clean — the binding check; certificate rows enter as kernel-`decide` facts. (ii) `check_cert.py`: ACCEPT for both certificates (exact integers, tiling of $[0, 4{\cdot}10^5]$). (iii) Independent synthesis re-sweep (`verify_central.py`, fresh implementation): max $\psi_L + \psi_H - 1 - \tau = 1000$ (argmax tie 2647/2648) and 1529 (argmax 3559), witness tuples reproduced digit-exact; tail facts re-verified at $\tau \in \{4 \cdot 10^5{+}1, 10^6, 10^7, 10^9\}$. (iv) Mutation tests: corrupted certificates rejected. Fresh re-runs of (ii)–(iii): `numerics_c3p4_fresh.log` (C3-P4 lens).

6. Theorem 1.3: the eventual value $h_4(n) = 4$ for $n \geq 331{,}777$ [K]

Positive variant, modern convention ([vD26]/Lean `hFun`); the historical [Er72] phrasing demanded distinct *sums* (A6), so we claim the value of h_4, not the verbatim 1972 constant. Lower half = Thm 3.1 [**K**]. Upper half [**K**]: for $n \geq N_0 := 331{,}777 = 24^4 + 1$, every $A \subseteq [1, 2n]$ with $|A| = n + 4$ has an (h,4)-config. The whole theorem is verified by the Lean 4 kernel:

> `Erdos866.h4_eq_4 : ∀ n, 331777 ≤ n → hFun 4 n = 4` (module `Erdos866/H4Dichotomy.lean`, imported by the sorry-free root; axioms exactly `[propext, Classical.choice, Quot.sound]`; audits `auditC4P1.log`, `auditC4Synth.log`; in-file corollaries `h4_eq_4_charter` and `h4_eq_4_c2` pin the superseded paper-grade constants 4,593,324 and $8 \cdot 10^7$.)

The hand proof evolved in three rungs — $8 \cdot 10^7$ (`lenses/P4-frontier/DICHOTOMY.md`), 4,593,324 (the KC1 = 2 kill-count correction, `lenses/C3P3-frontier-transfer/GTRANSFER.md` §5), and the kernel chain below (the per-o-pair balanced-window collapse, C4-P1), which is what is formalized; structure:

Setup. D = missing odds, $d = |D|$, so $|E| = d + 4$ for the even part E. Two constructive machines: the **(2+2) machine** (two even anchors $s, t \in E$, an odd pair summing to s, an even pair summing to t; the four odd cross sums must all be hit by D) and the **triangle machine** (a *positive triangle* $2g_1 < 2g_2 < 2g_3 \in E$ with $g_1 + g_2 \geq g_3 + 1$ yields $x < y < z$ same-parity with $x{+}y, x{+}z, y{+}z \in A$; sweeping the opposite-parity fourth point w forces **(T⋆)**: every positive triangle has $z \geq 2n - 2 - 6d$).

The kill-count (K), per-o-pair. Per fixed (s, t, o-pair), each missing odd $m \in D$ kills at most **2** e-pair combos: m blocks an e-pair only via a cross sum $m = o_i + e$, so with the o-pair (o_1, o_2) fixed there are at most two candidate even values $e \in \{m - o_1,\ m - o_2\}$, and each determines its partner $t - e$, hence the whole e-pair (`K_epairs_le2`; the bound 2 is tight — verified exhaustively on parameter grids, `killcount_check.py` KC1 and the independent grid in `verify_constants.py`). The original write-up used the (true but not tight) aggregate bound 4.

The balanced-window collapse (`balanced_kill`). The per-o-pair form is strictly stronger than the aggregate use: with one *balanced* o-pair of s fixed, the window of balanced e-pairs of t (e_2 even in $(t/2,\ t/2 + 4d + 2]$) hosts $2d + 1$ candidate e-pairs, while the fixed o-pair leaves at most $2|D| = 2d$ of them killable. One surviving e-pair gives the (2+2) machine its four cross sums — contradiction. This single lemma replaces *both* zone lemmas of the earlier chains (the middle exclusion X1/X1′ and the triangular-grid X2/X2′), at validity $n \geq 8d + 8$ (slope 1/8, halving the earlier $16d$):

- **MID** ($s = t$): $E \cap [8d + 8,\ 2n - 4d - 5]$ is empty;
- **LOW×TOP** ($s \in [4, 8d + 6]$, $t \in$ TOP): impossible, so LOW $\subseteq \{2\}$;
- **TT**: three TOP elements give a positive triangle with $z \leq n + 2d + 1$, killed by (T⋆) ($3d + 1$ anchors $w = 2j - (1 - x \bmod 2)$, kill map $(m, \mathrm{pos} \in \{0, 1, 2\})$), so $|\mathrm{TOP}| \leq 2$.

If TOP $\neq \emptyset$: $|E| \leq 1{+}2 = 3 < d{+}4$ — contradiction. If TOP $= \emptyset$: (T⋆) makes $\Lambda := E/2 \subseteq [1, 4d{+}3]$

triangle-free ($\lambda_{i-2} + \lambda_{i-1} \le \lambda_i$), so $\max \Lambda \ge F_{|\Lambda|+1} \ge F_{d+5} > 4d + 3$ — **false for every** $d \ge 0$ (`triangle_free_fib` + `cap_fib`; checked to $d = 1000$ in exact arithmetic as belt). The Fibonacci capacity bound (CAP) thus closes the structural regime outright for all $d \ge 0$: no finite endgame enumeration occurs anywhere on the proof path. ■

Dense regime ($d > (n-7)/8$). Then $|E| = d + 4 > (n-7)/8 + 4 \ge F_4(2n-2)$ for $n \ge 331{,}777$, and upstream **Lean-verified** `ceslemgeneral_pos` ($k = 4$) applied to E produces four distinct positive integers with all pairwise sums in E. This is the only non-elementary ingredient. The crossover is certified in exact arithmetic and in the kernel: with $n = u^4$-scaling, the threshold is the integer root bracket $q(23) < 0 < q(24)$, i.e. $u \ge 24$, giving $N_0 = 24^4 + 1 = 331{,}777$; in $s = u - 24$ the majorization polynomial is $3s^4 + 220s^3 + 5404s^2 + 44952s + 12162$, all coefficients positive (identity + threshold re-checked by `verify_constants.py`, ALL PASS, re-run at C4 synthesis). N_0 is *not* optimized (the $u \ge 24$ threshold is for this majorization; $\sim 2.4 \cdot 10^5$ is plausible) — exact-value work supersedes constant-shaving. ■

The earlier chains as belt. The first proof of Thm 1.3 (C2, with aggregate (K) ≤ 4) ran the dichotomy at $n \ge 32d + 10$: there CAP only forces $d \in \{1, 2, 3, 4\}$, leaving exactly 92 triangle-free structures, each killed by exhaustive search — machine-checked twice with independent codebases (`endgame.py` branch-and-bound; `endgame_brute.py` full enumeration with an independent combo generator): RESIDUAL EMPTY; its exact-rational crossover is pinned at 70,192,954. The KC1 = 2 aggregate chain (GTRANSFER §5) then gave $N_0 = 4{,}593{,}324$ at validity $16d + 8$, with its own self-asserting exact-rational certificate (`n0_v2.py`: success at 4,593,324, failure at 4,593,323). Everything in both earlier chains remains verified — the 92-structure packing certificates stay kernel-closed in the module (`decide`-checked) even though they are now off the proof path — and they are retained as redundant belt.

Sharpness. A_n^* sits exactly on every machine boundary: LOW = $\{2\}$ (no o-pair), $|\mathrm{TOP}| = 2$, and its only mixed triple $(1, n-1, n)$ has $g_1 + g_2 = g_3$ — the $x = 0$ escape. Surplus 4 leaves no room.

Honest accounting. The entire theorem — structural reduction, zone collapse, CAP, dense regime, crossover, and assembly — is now verified by the Lean 4 kernel (`h4_eq_4`); the former trust locus (the hand-proved structural reduction) is discharged. Machine checks beyond the kernel: the kill-count constants (exhaustive grids, `killcount_check.py` and the independent grid in `verify_constants.py` — worst case exactly 2, tight), the crossover identity and threshold (exact-rational, `verify_constants.py` ALL PASS), 20k+20k random-instance checks of both balanced-window constructions, the validity-collection sweep to $d \le 9999$, the belt-chain finite endgame (twice, independent codebases), and empirical validation of the machines (`validate.py` ALL PASS: no false fire on 120 lab extremal sets; fires on $A_n^* + x$ for every x, $6 \le n \le 120$; 400 adversarial random instances; greedy-blocked near-residual structures). Hostile Codex reviews: the original chain (thread 019ebb83…) ACCEPT-WITH-EDITS, all edits applied; the kill-count correction and $N_0 = 4{,}593{,}324$ (thread 019ebd11…) ACCEPT-WITH-EDITS; the kernel port and per-o-pair collapse (thread 019ebd9f…) ACCEPT-WITH-EDITS — "I did not find a hidden weakening" — with the foil independently re-running audit + constants + full build. The gap between the last fully-chained exact cell (§9.2) and N_0 stays at $h_4 \in [4, 1000]$ (Thm 1.1) and is closable only by a linear-density dense lemma (open, §11); SAT cannot reach it.

7. Theorem 1.4: the forbidden-set recursion and $g_5(n) \le 3{,}519{,}219$

g-side. Define $f_3^*(x, \varphi) := \varphi + \frac{1}{2} + \sqrt{x + 1 + (\varphi - \frac{1}{2})^2}$ and $f_k^*(x, \varphi) := \varphi + \frac{1}{2} + \sqrt{x\, f_{k-1}^*(x, \varphi+1) + (\varphi - \frac{1}{2})^2}$.

Lemma 7.1 (the new lemma; replaces [vD26] Lemma 7). Let $k \geq 3$, F a finite set of positive integers, $|F| = \varphi$, and $A_0 = \{a_1 < \cdots < a_r\}$ a set of even integers ≥ 2 with diameter $x := a_r - a_1 \geq 2$ (i.e. $r \geq 2$; the Lean statement carries the range guard explicitly — see §7.3). If $r \geq f_k^*(x, \varphi)$ there exist integers $c_1, \ldots, c_k$ with (1) $c_1 \in A_0$; (2) $c_2, \ldots, c_k$ nonzero, pairwise distinct, none in F; (3) every subset sum containing c_1 lies in A_0.

Proof idea. Induction via $U_m := A_0 \cap (A_0 - m)$ with forbidden set $F \cup \{m\}$. Van Doorn's Lemma 7 demands s *disjoint* representations of m — costing a factor 2 per recursion level — only to force $m \neq c_i$; the forbidden set does that bookkeeping at additive cost $\varphi(r-1)$ in the count instead. Base $k = 3$: if every allowed m had $|U_m| \leq 1$ then $r(r-1)/2 \leq \varphi(r-1) + x/2$, contradicting $(r-1)(r-2\varphi) \geq x+1$; an allowed m with $u < u' \in U_m$ gives $(c_1, c_2, c_3) = (u', u-u', m)$. Step: some allowed m has $|U_m| \geq (r-1)(r-2\varphi)/x \geq f_{k-1}^*(x, \varphi+1)$; recurse on U_m, append $c_k := m$. ■

Corollary 7.2 ($\varphi = 0$). $r \geq f_k^*(x, 0) \Rightarrow A_0$ contains a (g,k)-config ($b_1 = c_1/2$, $b_i = c_1/2 + c_i$).

Theorem 7.3 (= 1.4, [K]). $g_5(n) \leq C_5 = 3{,}519{,}219$ for all $n \geq 1$: van Doorn's Theorem 8 proof verbatim with Lemma 7 $\to$ Cor 7.2 (φ-chain $f_5^*(x,0) \to f_4^*(x,1) \to f_3^*(x,2)$), where C_5 is the least integer with $x/12 + C_5/2 - 2 > f_5^*(x, 0)$ for all $x \geq 1$ (**xineq***). The case split (slope 1/12, central window, kill-rate 3) is unchanged. Improvement factor over the paper-internal constant: 32.3× (113,591,719/3,519,219); over the Lean-stated $1.2 \cdot 10^8$: 34.1×.

Why only factor 32 and not more (negative results, [P]). The audit of every lossy step shows S2 (disjointness) was the *only* first-order slack: slope 1/12 is rigid in this skeleton (both end-margins forced $\geq 6\tau + 2$; kill-rate 3 tight), asymmetric windows buy nothing for g_5, and the new recursion is *worse* for the single-level h_4 chain. Dead-end map preserved deliberately (referee value; `new-recursion.md` §4). An "O'Bryant-with-exclusions" base would push C_5 to $\approx 1.81 \cdot 10^6$ — open (§11).

Verification. (i) Lean kernel — the binding check: `g5upper_star_charter : ∀ n, gFun 5 n < 3519220` sorry-free, axioms clean (§7.3). (ii) (xineq*) and minimality of C_5: pure-integer interval certificate (`certify.py`, isqrt outward rounding, scale 2^{48}): 196,759 monotonicity intervals tile $[1, 10^{16}]$ (min slack 0.0745 at $x \approx 1.4777 \cdot 10^8$), decreasing-ratio tail for $x \geq 10^{16}$, $C_5 - 1$ violated at $x = 147{,}760{,}000$; mutation tests rejected. (iii) Independent synthesis re-implementation (Decimal-60) matches the local minimum to 10 significant digits. (iv) Independent exact-arithmetic re-verification of the Lean numeric layer (`verify_numerics.py`: tangency identities exact in value and slope, slope gap $6.6 \cdot 10^{-7} > 0$, constant margin 14.25). Gate-2 anchors: the same code path reproduces van Doorn's 113,591,719 and the h_4-sketch constant 3166 exactly. Fresh re-runs: `numerics_c3p4_fresh.log`.

7.3 Lean port (complete) and the statement repair

The port is **complete and kernel-verified at the charter constant**: `Erdos866.g5upper_star_charter (n) : gFun 5 n < 3519220` and the pinned target `T1TargetG5_closed_charter`, sorry-free on the three base axioms, root-imported, fresh build green (34.1× below the upstream Lean constant `g5upper`'s $1.2 \cdot 10^8$; audits §1.3). Ingredients: the Lemma 7.1 induction is fully proved (`lemmaA`, by `Nat.le_induction` over k, at **general** $k \geq 3$ — the disjoint-representations bookkeeping genuinely disappears: distinctness of the appended element comes from $b_i \notin F \cup \{m\}$), together with the general-k extraction lemma `ceslemgeneral_star`; van Doorn's Theorem-8 case tree is formalized once, parameterized over the constant c with a single key-inequality hypothesis, and instantiated at $c = 3{,}519{,}219$. The numeric layer avoids any 196k-row certificate replay: the

$u := x^{1/8}$ substitution turns the chain into square-root-free polynomial inequalities (`key_ineq_star`, `key_ineq_star_tuned`, and the charter `key_ineq_star_charter` via a two-regime split at $u_c = 52/5$ whose sharp-regime margins — slope gap $3.5 \cdot 10^{-12}$, constant margin 0.016 — are razor-thin but exact-rational, checked by the kernel). Two hostile Codex passes on the port (one per rung) returned ACCEPT, the second with zero findings.

Formalization already paid off mathematically: as prose, Lemma 7.1 was **false for singletons** ($x = 0$: $f_4^*(0,0) = 1 = |A_0|$ with no structure) — the Lean statement carries the range guard `2 ≤ max - min`, and the prose above is the repaired form. A second trap surfaced by the port: the parameterized Theorem-8 wrapper needs $c \geq 7$ (not $c \geq 4$) for its concentrated-edge pigeonhole — harmless at $c = 3{,}519{,}219$ but a real failure mode for anyone re-instantiating the wrapper at tiny constants. $C_5 = 3{,}519{,}219$ is the certified minimum for this route (Prop.-5.4-style route-optimality): going lower needs a new route, not tuning.

8. General k — Theorem 1.7 [P]

For every $k \geq 4$ and $n \geq N(k) := k^4 \cdot 2^{4k+27}$ (explicit, not optimized): $g_k(n) < 2\,n^{1-2^{2-k}}$ and $h_k(n) < 2.25\,n^{1-2^{2-k}}$, improving [vD26 Thm 9]'s constant 4 uniformly with an explicit threshold; the per-k limsup constants are $\gamma_k = 2^{1-2^{2-k}} \uparrow 2$ (g-side) and $\mu_k = 2((k-1)/2)^{2^{2-k}} \downarrow 2$ (h-side, $k \geq 4$). The g-chain runs van Doorn's argument through the forbidden-set recursion f^* of §7 at general k (lead constant 1, was 2); the h-chain needs one new ingredient, **Lemma A′** (a base swap for the h-side: counting-with-exclusions $|U_m| \geq \varphi + 3$ forces all-positive c_i; its mechanics mirror the kernel-checked Lemma 7.1). Full write-out, proofs, and constant reconciliation against the kernel $k = 4, 5$ closures (no tension; the T1 theorems dominate at those k): `lenses/C4P3-allk-writeup/T2-ALLK.md`. Machine verification: `verify_allk.py`, exact Fraction/isqrt interval arithmetic, **272 checks PASS** (re-run at the C4 synthesis), including recursion numerics at $k = 6, 7, 8$ and endgame margins $\geq 1.08 \cdot 10^{-2}$; the Lemma A′ base case was additionally re-verified by an independent brute-force implementation over 106 exhaustive + 16,224 random hypothesis-meeting instances (`lab/synthesis_c4/`). Formal anchors (`lemmaA`, `ceslemgeneral_star`, both T1 closures) are kernel-grade; the all-k statement itself carries **no Lean object** (honest [P]; a `gFun k n` wrapper on `ceslemgeneral_star` is the cheapest formalization path, C5-P3).

9. Exact values, extremal structure, conjectures — [E]

9.1 Method and certificate standard

Values $m_k^v(n)$ (max config-free size) by CEGAR over CaDiCaL with a definition-direct witness oracle. Every finished cell carries: (i) extremal sets re-checked by two independent config checkers (Python DFS; Rust brute force); (ii) clause-tuple UNSAT certificate re-proved by two engines from different codebases (z3; CaDiCaL on a sequential-counter encoding); (iii) an archived **DRAT/LRAT proof**: reference-CaDiCaL DRAT, `drat-trim` "s VERIFIED" (emitting LRAT), `lrat-check`, and finally **cake_lpr** — a formally verified (CakeML) checker — "s VERIFIED UNSAT". Per-cell artifacts and sha256 in `sat_archive/manifest.json` (quote the manifest at submission, not any prose count). The archive verifies CNF-level UNSAT; encoder semantics rest on exhaustive SAT-free cross-checks of 14 tiny cells (no shared code) plus mutation controls (truncated DRAT, deleted clause, flipped literal — all rejected). The [E] grade requires both the lab cell and a manifest entry that is either **VERIFIED** or **TRIVIAL_M_EQ_N** — the latter for cells with $m_k^v(n) = 2n$ (the whole window is config-free), where no UNSAT proof exists *by design* and the value is checked by direct construction; the dual-engine + DRAT/LRAT chain applies to the nontrivial cells only. One archive incident is disclosed for the record: the $h_4(62)$ cell's first DRAT proof was *rejected* by drat-trim's backward RAT check (deletion-line mismatches; the 382 MB proof was emitted while five other SAT jobs saturated the machine, so transient proof-file corruption under I/O load is the leading explanation). Not a soundness event — a rejected proof certifies nothing and claims nothing — and the resolution kept the trust chain intact: a clean re-run of the *same* default solver on the identical CNF produced a proof that passed the full drat-trim → lrat-check → cake_lpr chain (manifest entry VERIFIED; the archive tool additionally gained a recorded `cadical --plain` fallback for any future rejection). A cell enters a claimed range only once its manifest entry is VERIFIED. Full-lab dual-engine re-verification: the canonical `lab/data/verify_report.json` covers the frozen 298-cell snapshot (every cell with a claimed value; the cell list is read from the pinned SAT-archive manifest, sha256 `45f0eedc...`, never from a directory glob) with the identical per-cell check throughout — `verify_cert.py --both` (z3 **and** CaDiCaL UNSAT at $|A| \geq M+1$). For wall-clock reasons the pass was assembled from several runs rather than one process — a detached cold pass (2026-07-08, 90 cells; transcript `verify_canonical_c5.log`), four stride-sharded resume runs over the remainder, and a cloud race whose per-cell results were accepted only after a local re-hash of each cell's input against the record and a strict match of the dual-engine output format — and the report discloses this: it carries `resumed: true`, the manifest sha256, and a `runs` list naming every contributing transcript with its cell count. **Final counts: 298/298 cells, 0 failures** (completed 2026-07-08; SUBMIT.md gate G1 GREEN; also quoted in Appendix B). Cells landing after the snapshot ($h_4(65)$–$h_4(68)$, $g_5(24)$, $g_5(25)$) are not covered and not claimed.

9.2 Exact-value tables

Generated from cells by `make_tables.py` (authoritative copy: `lenses/C4P4-publication/paper/tables-866.md`, regenerated at this draft; do not hand-edit). Headlines:

- $g_5(n) = 4$ **for** $15 \leq n \leq 23$ — the first exact g_5 values beyond van Doorn's $n \leq 15$ search, directly on his flagged question ("cannot even exclude $g_5(N) \leq 4$ for all large N"); $g_5 = 5$ on $5 \leq n \leq 14$ (sporadic onset). Every cell in the range carries DRAT/LRAT + cake_lpr VERIFIED archive proofs and Rust-checked extremal sets; the $n = 22$ value is exact with extremal-set enumeration time-capped (7 extremal sets recorded), and $n = 23$ is a full exact solve (64 extremal sets, all Rust-checked).
- $h_4(n) = 4$ for $n \in \{3,4\} \cup [6,64]$; $h_4(5) = 5$ (sporadic); with **unique** extremum A_n^* for $11 \leq n \leq 64$ (enum-exact; equality with A_n^* checked cell-by-cell; the 62–64 archive chains landed at this draft — see §9.1 for the disclosed $h_4(62)$ proof-rejection incident and its clean-re-run resolution). The dual-engine (z3 + CaDiCaL) re-proof of every cell, including these, is the release-gated canonical pass of §9.1 (SUBMIT.md G1).
- $h_5(n) = z(n) + 1$ for all computed $13 \leq n \leq 26$, spanning the Fibonacci jump at $n = 21$.
- $g_4(n) = 3$ at every computed $n \geq 2$ (through $n = 40$, plus the $n = 60$ cell landed at this draft), with unique extremum = van Doorn's family odds $\cup \{2n-2, 2n\}$ at both $n = 40$ and $n = 60$ — direct exact refutations of the agentic-erdos $g_4(40)/g_4(60) \geq 4$ claims (§9.4).

9.3 Conjecture ledger (tested against every computed cell)

(C1) $h_4(n) = 4$ for all $n \geq 6$ — settled for $n \geq 331{,}777$ (Thm 1.3, kernel-grade) and at the computed exact cells (§9.2); open in between (a finite but SAT-infeasible gap; see §11, item 2). (C2) $g_5(n) = 4$ for all $n \geq 15$ — the g-side transfer question; the $x = 0$ equality-triangle escape is legal on the g-side, so the (T⋆) machine *strengthens*, and the endgame becomes a (3,2)-machine enumeration with a zero anchor. Status: **the structural program is now §10** — the per-anchor dichotomy (†)

[P], Theorem Z's all-low/all-top zone reduction ([P] minus one named audit), a conditional capacity ceiling shrinking the open band to $d \in [5, 19]$, and an in-flight n-free enumeration, **empty in every completed line** (m=4 through $d \leq 5$ (L) / $d \leq 4$ (T), m=5 through $d \leq 8$, plus the complete $d = 0$ case at $n = 15..55$ and 2009/2010) — in particular the A_n^*-type h-side extremal *dies* on the g-side, killed exactly by the legalized $x = 0$ anchor. Remaining before a theorem: §10.5's named list (sweep completion, Lemma C at general width, the $d = 0$ write-up, the Theorem-Z audit, the exact $N_{0,g}$ crossover). (C3) $h_5(n) = z(n) + 1$ for all $n \geq 13$.

9.4 Refutation note

The repository przchojecki/agentic-erdos claims randomized-search lower bounds $g_4(40) \gtrsim 4$ and $g_5(40) \gtrsim 13$. These contradict the Lean-verified $g_4(n) = 3$; the bug is reproducible — requiring $b_i \in A$ reproduces all four claimed cells exactly. Explicit witnesses (e.g. $b = (0, 1, 2, 77)$ at $n = 40$) and diagnosis: `lenses/P5-hygiene/refutation/refutation-przchojecki-866.md`. The $g_5 \approx 13$ claims carry no evidential weight (same broken checker), though they are not formally disproved.

10. Toward $g_5 = 4$: the per-anchor collapse — graded per part, no exact-value claim

This section reports the g-side structural program (C5-P1 lens, artifacts `lenses/C5P1-g5eq4/`: `GZONES.md`, `verify_gkill.py`, `verify_zones.py`, `endgame_v2.py`, `d0_check.py` + logs). **We do not claim $g_5(n) = 4$ or $g_5(n) \leq 5$ for any n here.** Grades are stated per part; none of this section is [K].

Conventions as in §6: $A \subseteq [1, 2n]$ config-free at surplus m, D the missing odds with $d = |D|$, $\Lambda = E/2 \subseteq [1, n]$ the halved evens, $p = |\Lambda| = d + m$. For a triple $T = \{a < b < c\} \subseteq \Lambda$ write $x = a + b - c$, $y = a + c - b$, $z = b + c - a$; on the g-side $x \leq 0$ is *legal* as the unique non-positive summand — exactly the equality-triangle escape that separates g_5 from h_5.

10.1 The unified per-anchor dichotomy (†) — [P]

The per-o-pair collapse that closed Thm 1.3 (§6, `balanced_kill`) transfers to the g-side and comes out *stronger*: the three-branch g-side case algebra (the (A)/(B)/(N) analysis of the earlier transfer notes, `GTRANSFER.md` §3) merges into one two-sided dichotomy per anchor. Three lemmas (proofs: `GZONES.md` §1):

- **Lemma U (window unification).** For fixed (T, u), $u \in \Lambda$, the f-pairs $(2u - f_2, f_2)$ completing (x, y, z) to a 5-config candidate are exactly those with $f_2 \equiv (a + b + c + 1) \bmod 2$ and $u < f_2 \leq \min(2u - 1 + x,\ 2n - 1 - z)$ — one contiguous window. The zero/negative anchor lives *inside* the window; the separate case
  (N) disappears.
- **Lemma K (per-anchor kill count).** One missing odd kills at most **3** admissible f-pairs of a fixed (T, u) (tight).
- **Lemma G (g-side balanced kill).** In a config-free A with odd coverage there is no (T, u) with $u + x \geq 6d + 4$ and $u + z \leq 2n - 6d - 4$: the $3d + 1$ stepped pairs beat the $3d$ kill budget. Hence, with $K_1 := 6d + 2$:

  (†) for every $T = \{a < b < c\} \subseteq \Lambda$ and every $u \in \Lambda$: $u + (a{+}b{-}c) \leq K_1 + 1$ **or** $u + (b{+}c{-}a) \geq 2n - K_1 - 1$.

Machine verification (`verify_gkill.py`, exact integers): window = definition *exhaustively* over 267,503 (n, T, u) cells ($n \in \{12, 13, 30, 31\}$); kill map worst count 3, tight, over 40,012 nonempty cells; 109,386 Lemma-G construction instances (admissible + distinct); 3,000 random (†)-violating (n, Λ, D) each completed to an explicit 5-config with all 10 sums verified in A. One hostile review pass (verdict ACCEPT on this part); independently re-run green at the C5 synthesis and by the reviewer. Grade: **[P]** (written proof + exhaustive machine checks at small n + independent re-runs). Lean port is chartered, not done: `G_kill` has the same shape as the kernel-verified `K_epairs_le2` of §6.

10.2 Theorem Z: zone reduction from (†) alone — [P] minus one named audit

Theorem Z. Let A be 5-config-free at surplus $m \geq 4$ with $p = d + m \geq 5$ and $n \geq 66d + 36$. Then $\Lambda = \{\lambda_1 < \cdots < \lambda_p\}$ is either **(L) all-low** — $\lambda_{p-2} + \lambda_{p-1} \leq K_1 + 1$ (so the $p - 2$ core elements lie in $[1, 3d + 1]$) and $\lambda_p \leq 12d + 6$ — or **(T)** the exact **all-top mirror** in offsets $\delta_i = n - \lambda_i$. In particular: no mixed low/top sets and no middle elements. (The 12d+8 two-sided windows of the earlier draft-grade reduction shrink to a $3d + 1$ core plus two stragglers, and the reduction is now a one-page case tree of (†) instances — `GZONES.md` §2.)

Machine verification (`verify_zones.py`): 59,228 stratified adversarial samples ($d \in [1, 8]$; mixed/mid/half-point/Fibonacci/ cluster shapes) — every (†)-consistent sample lies in (L) $\cup$ (T) — plus an exhaustive $d = 1$ boundary sweep. Hostile review verdict ACCEPT-WITH-EDITS; all findings adopted in place. Grade: **[P] minus one named pending item** — a fresh-eyes audit of the case tree (the arithmetic at the binding validity $n = 66d + 36$ was already once corrected under review), chartered for the next campaign. We do not build any public claim on Theorem Z until that audit lands.

10.3 Capacity ceiling — conditional [D]: the one mathematical hole is named

Lemma C (open at general width). If some sum S has $r(S) \geq 3$ representations $S = \alpha + \beta$ ($\alpha \leq \beta \in \Lambda$), then three of them admit a role-assignment forming a *valid 4-block* (4 distinct integers, ≤ 1 non-positive, all 6 pairwise sums even and present), which is fatal for $n \geq 16d + c_0$ (its anchor window carries $> 4d$ survivors at kill count ≤ 4). Evidence: exhaustive machine decision at widths $W \leq 18$ (21,817 $r \geq 3$ instances, zero failures) extended independently by the hostile reviewer to $W = 24$ (701,481 instances, zero failures); translation covariance moves fixed-width configurations but does **not** lift the evidence to arbitrary width. **The general-W case analysis is an open write-up; the reviewer's verdict on this part was REJECT-as-written until it is proven, and we adopt that**: everything downstream of Lemma C is stated conditionally.

Conditional on Lemma C, Theorem Z's core-in-a-box forces, by pair-sum counting in a box of $3d + 2$ values: at $m = 4$, $d \leq 18$ (L) / $d \leq 19$ (T); at $m = 5$, $d \leq 16$ (L) / $d \leq 17$ (T). The untouched middle band of the earlier analysis, $d \in [5, \sim 87]$, shrinks to $d \in [5, 19]$. Grade: **[D], conditional** (machine evidence exhaustive at small scales; general proof open).

10.4 The middle band as a finite n-free enumeration — [E]-in-progress

For zone candidates the (†)-window endpoints resolve n-freely in mirror/offset coordinates (given n's parity), so each $(m, d, \text{branch}, n\text{-parity})$ candidate set and its hitting-set verdict is independent of n beyond the validity threshold — this *removes* the n-stability scope limitation of the earlier census. `endgame_v2.py` enumerates all (L)/(T) candidates with +2 slack boxes at four large n (two per parity) and asserts equality of normalized per-candidate verdict maps across same-parity runs (upgraded from raw counts under review). DFS pruning (per-(T, u) counts; valid 4-blocks) is sound under extension.

Sweep status at this draft (disk-read 2026-06-12 21:33 machine-local; detached jobs still running): every completed line **EMPTY** with the n-free certificate OK — $m = 4$: (L) through $d \le 6$, (T) through $d \le 5$; $m = 5$: both branches through $d \le 8$; **0 survivors in all 31 completed** $(m, d,$ **branch**$)$ **lines**; targets are the conditional ceilings of §10.3 ($d \le 19$ resp. 17). The $d \le 4$ lines reproduce the independent C3-era enumeration exactly. Separately, the $p = 4$ case ($d = 0$, $m = 4$ — outside Theorem Z's $p \ge 5$): exhaustive over **all** Λ (no zone assumption) for $n = 15..55$ and wide-window sweeps at $n = 2009/2010$, all EMPTY, run **DONE**; its O(1) hand write-up is a named TODO and stays out of any assembly. Grade: **[E]-in-progress** (finite computation; harvest incomplete — any survivor line would itself be a discovery: the first concrete g-side obstruction family beyond the h-side extremal).

10.5 Assembly shape and honest status

If (and only if) the named items close — sweeps empty through the ceilings, Lemma C at general W, the $d = 0$ write-up, the Theorem-Z audit — then $g_5(n) = 4$ assembles for $n \ge N_{0,g}$ as STRUCTURAL (Z + enumeration $d \le 19$ + capacity $d \ge 20$) + DENSE (upstream `ceslemgeneral`, $k = 5$) with $N_{0,g} \approx 10^{18}$ at the proven validity slope 1/66; the slope optimization (66 $\to\sim$ 16, tightening the threshold chain) is the named lever back toward $\sim 2 \cdot 10^{13}$. The $g_5 \le 5$ rung has the identical skeleton at $m = 5$ and ships only when the same machinery closes there. Until then, the strongest true statements remain Thm 1.6's exact cells ($g_5(n) = 4$ on $15 \le n \le 23$) and the (†)/Theorem-Z structure above at their stated grades.

11. Open problems

1. $g_5(n) = 4$ **for all** $n \ge 15$**?** (van Doorn's question; the actual Erdős-side ask.) The per-o-pair collapse of §6 *does* transfer — §10 derives the one-window per-anchor dichotomy (†), Theorem Z's two-shape zone reduction, and a conditional capacity ceiling at $d \le 19$, with the n-free enumeration empty in every completed line. What remains is §10.5's named list, headed by Lemma C at general width and the sweep harvest; assembly would give $g_5(n) = 4$ for $n \ge N_{0,g} \approx 10^{18}$ at slope 1/66, with the slope optimization ($\to\sim$ 16) the lever back toward $\sim 2 \cdot 10^{13}$. The first rung $g_5 \le 5$ (large n) has the identical skeleton at $m = 5$.
2. **The gap** $[65, N_0)$ **for** $h_4 = 4$**:** needs a dense lemma at linear density (F_4-type bounds are $n^{3/4}$); further constant-tuning of $N_0 = 331{,}777$ within the current chain is not expected to beat $\sim 2.4 \cdot 10^5$. SAT can never reach N_0; ladder extension only nibbles the left end.
3. **O'Bryant-with-forbidden-differences:** would push C_5 to $\approx 1.81 \cdot 10^6$ and flip the h_4 chain.
4. **Slope-1/12 barrier** for g_5: provably rigid in the vD skeleton (§7); new geometry needed for orders-of-magnitude progress.
5. **g-version of CES Theorem 6** (ambiguity A9): is $g_k(n) > n^{1-\epsilon}$ for $k \ge k_0(\epsilon)$? The CES example analysis predates the g/h split.

12. Methods, verification doctrine, and AI disclosure

AI disclosure. The mathematics in this paper was produced by an AI-agent workflow: Claude (Anthropic) agents generated the proofs, certificates, code and Lean formalizations across three campaigns of an iterated attack program; GPT-5.5 (OpenAI, via Codex) served as a standing *adversarial* referee — every headline result passed at least one hostile review pass (verdicts and finding dispositions are archived with thread ids in the program records), and review findings led to real repairs (notably the Lemma 7.1 singleton bug, §7.3, and the fixed-(s,t) qualifier in §6's (K)). The upstream Lean development we build on was formalized by Aristotle (Harmonic) for van Doorn [L866]. Human role: program direction and final review. Every theorem stated as [K] is machine-checked by the Lean 4 kernel and carries an axiom audit. Both previously named trust loci are now **discharged in Lean**: the hand-proved structural reduction of §6 (Thm 1.3's upper half) closed kernel-grade as `structural_case/h4_eq_4` (C4-P1; the (K) double-count scaffold was formalized *first*, precisely where a hand error would have surfaced, and the reduction survived), and the Lemma 7.1 induction of §7 was fully proved earlier (§7.3), the formalization process itself catching and repairing a real statement bug (the singleton case). The remaining non-kernel claims are graded [P]/[E] with their checking artifacts named (§8, §9). No [K] label in this paper rests on unverified AI output; [P]/[E]/[D] labels state exactly which parts are machine-checked and which are reviewed prose.

Verification doctrine. (i) Exact arithmetic only — integer/rational interval certificates with outward rounding; no floating-point claim enters a proof. (ii) Dual independent checkers for every computation (different codebases, ideally different languages/engines). (iii) Negative controls: mutated certificates must be rejected. (iv) Independent re-verification at each campaign synthesis (fresh implementations, not reruns). (v) SAT results carry archived DRAT/LRAT proofs checked by a formally verified checker (§9.1). (vi) Lean: upstream byte-pinned (sha256-verified at every audit), extension modules sorry-free, `#print axioms` on every cited theorem, fresh `lake build` at audit time. (vii) An openness kill-check against the literature is re-run immediately before any public claim (§13).

Reproducibility. Single entry point per claim; all spawned compute at idle priority; artifact map in Appendix B.

13. Openness kill-check — FRESH RUN AT DRAFT COMPLETION

Run 2026-07-08 (release freeze, Draft v5) — all 8 items re-fetched live this session. erdosproblems.com/866 HTTP 200 (browser-grade fetch), OPEN, "cannot be resolved with a finite computation", last edited 01 December 2025, still citing $g_4 \leq 2032$; forum thread 866 newest comments still 04 May 2026; arXiv:2605.00040 still **v1 only** (live fetch of the submission history); Woett `ErdosProblem866.lean` still `1e075c4f6e8a` (2026-04-30, GitHub API); przchojecki still `c58f8589` (2026-05-10); teorth problems.yaml #866 `state: "open" / formalized: "no"` (raw, live); web search (multiple phrasings) clean. **One delta, assessed non-superseding:** plby/lean-proofs has continued work on its `Erdos866b.lean` port (2026-06-29 v4.32.0 re-add, 2026-06-29 build fix, 2026-07-02 proof fill-in; file at HEAD `97957fb9` is sorry-free) — the diffs were read: it formalizes the **same known theorem set at the same constants** ($h_4 \leq 2270$, $g_4 = 3$, $g_3 = 1$, van Doorn's results), no improved bound, no exact-value theorem; every bound this paper improves remains the published best. Verdict **GREEN**. Prior records retained for provenance:

Run 2026-06-13 (C6 session, Draft v4), by the C6-P2 release-gate lens — all 8 items re-fetched live this session: UNCHANGED on every item. erdosproblems.com/866 HTTP 200, OPEN, "cannot be resolved with a finite computation", last edited 01 December 2025, still citing 2032; forum newest 04 May 2026; arXiv:2605.00040 still v1 (live fetch); Woett `ErdosProblem866.lean` still `1e075c4f6e8a` (2026-04-30, gh API); plby/lean-proofs `6d0ba683` (2026-06-11, style-only); przchojecki `c58f8589` (2026-05-10); teorth problems.yaml #866 `state: "open" / formalized: "no"` (raw, live); web search clean (nothing citing, improving, or superseding). Verdict

GREEN. The C4-era record below is retained for provenance:

Run 2026-06-12 (C4 session), by the C4-P4 lens (this draft's author agent), per PROBLEM.md §8 — all items re-fetched live this session:

1. **erdosproblems.com/866** (fetched live, HTTP 200): status **OPEN** — "open, and cannot be resolved with a finite computation."; page last edited 01 December 2025 (unchanged since the freeze; still citing $g_4 \le 2032$).
2. **Forum thread 866**: newest comment still 04 May 2026 (van Doorn's bounds + Sothanaphan congratulations); nothing newer.
3. **arXiv:2605.00040**: still **v1 only** (submitted 2026-04-28); no revision, no superseding submission found by search.
4. **Woett/Lean-files** `ErdosProblem866.lean`: last commit touching the file still `1e075c4f6e8a` (2026-04-30) — GitHub API, live.
5. **plby/lean-proofs** `Erdos866b.lean`: last commit touching the repo 2026-06-11 ("Address linter.style.refine in Erdos866b") — style-only churn, same theorems, same constants.
6. **przchojecki/agentic-erdos**: last repo commit still 2026-05-10 (`c58f858`); no new claims.
7. **teorth/erdosproblems** `problems.yaml` (main, raw, live): #866 `state: "open"` (last_update 2025-08-31), `formalized: "no"`.
8. **Web search** (multiple phrasings): no citing, improving, or superseding work found.

Verdict: open; every standing bound this paper improves is still the published best. A strict-improvement claim dies if superseded; the formalization claims survive any scoop (the Lean theorems are ours regardless; the kill-check decides framing, not whether to finish). Re-run again immediately before submission.

Appendix A. Certificate file formats

`cert_energy.csv` / `cert_ruzsa.csv`: rows $(\tau_{lo}, \tau_{hi}, \Phi_L, \nu_L, \psi_L, \Phi_H, \nu_H, \psi_H)$ tiling $[0, 4 \cdot 10^5]$; conditions V2–V5 checked per row in exact integers (and, for the energy form, re-proved row-wise by Lean kernel `decide`). Cell JSON schema: `{n, k, variant, M, value, extremal_sets, clause_tuples, status, engines, time_s}`. DRAT/LRAT archive layout and the full soundness chain: header of `sat_archive/make_drat_archive.py`; per-cell sha256 in `manifest.json`.

Appendix B. Verification artifact map

Claim	Primary artifact	Independent check	Fresh re-run (this draft)
Thm 1.1	Lean `h4_le_1000` + `cert_energy.csv`	`check_cert.py` ACCEPT; `verify_central.py` re-sweep; mutation tests	`auditC4P4.log`; `numerics_c3p4_fresh.log` [1,3]
Thm 1.1′	`cert_ruzsa.csv`	same pair	`numerics_c3p4_fresh.log` [2,3]
Thm 1.2	Lean `four_le_hFun_four`	Python DFS + Rust checker on A_n^*	`auditC4P4.log`

Claim	Primary artifact	Independent check	Fresh re-run (this draft)
Thm 1.3	**Lean `h4_eq_4`** (H4Dichotomy, root-imported)	`verify_constants.py` ALL PASS (x24 identity, $q(23) < 0 < q(24)$, fib CAP to $d = 1000$, independent KC1 $= 2$ grid tight, 20k+20k balanced-window instances, validity sweep $d \leq 9999$; re-run at C4 synthesis); hand chains: DICHOTOMY.md + GTRANSFER.md §5 with `killcount_check.py`, `n0_v2.py`, belt `endgame.py`/`endgame_brute.py`, `validate.py`	`auditC4P1.log` + `c4p1_fullbuild.log` (8039 jobs); `auditC4Synth.log` + from-scratch `c4synth_build.log`; belt re-runs `n0_v2_rerun_c4.log` (star6 PASS at 4,593,324 / fail 4,593,323) + `killcount_rerun_c4.log` (KC1–KC4 ALL PASS)
Thm 1.4	Lean `g5upper_star_charter` + `certify.py` certificate	`verify_recursion.py` (Decimal-60); `verify_numerics.py` (exact tangency identities); mutation report	`auditC4P4.log`; `numerics_c3p4_fresh.log` [4,6]
Thm 1.5	Lean `fibCnt_lt_hFun_five`	`verify_families.py` DFS; 10^6 sweep	`auditC4P4.log`; `numerics_c3p4_fresh.log` [5]

Claim	Primary artifact	Independent check	Fresh re-run (this draft)
Thm 1.6	lab cells	dual-engine + DRAT/LRAT + cake_lpr archive	canonical `lab/data/verify_report.json` — frozen 298-cell snapshot, pinned-manifest cell list (sha256 `45f0eedc...`), per-cell `verify_cert.py --both`; **completed 2026-07-08: n_cells = 298, n_failed = 0** (SUBMIT.md G1 GREEN; multi-run assembly disclosed in the report's `runs` list — see §9.1); archive manifest 273 VERIFIED + 25 TRIVIAL_M_EQ_N = 298/298, 0 FAIL (G2); belt dual-engine re-proof of the late cells h_4(62–64), g_5(23): 4/4, 0 failures (`verify_new_cells_c5_report.json` G3)
§10 (g-side program; no claim)	`lenses/C5P1-g5eq4/GZONES.md`	`verify_gkill.py` (267,503-cell exhaustive window check; KC tight at 3; 109,386 + 3,000 instances), `verify_zones.py` (59,228 adversarial + exhaustive d=1; Lemma C 21,817 + independent 701,481 instances), both re-run independently at the C5 synthesis and by the hostile reviewer	sweep logs `eg_m{4,5}_{L,T}.log`, `d0_check.log` (in flight at this draft; status as quoted in §10.4, disk-read 2026-06-12 21:33 machine-local)

Appendix C. Lean formalization inventory (audited 2026-06-12/13 (C4–C5 sessions))

Toolchain: lean4 v4.28.0, mathlib 8f9d9cff6bd7. Upstream `Erdos866/Upstream.lean` byte-pinned, sha256 `043731e35444d446ad8effeb8a5f1febdc904991bf9b9312c3acc2c17ca9845e` (= GitHub Woett/Lean-files main @ 1e075c4f6e8a), re-hashed at the C4-P1 and C4-synthesis audits (Codex re-hashed independently). Root module imports (**all sorry-free**): Statements, Warmup, G5Small, LowerBounds, H4Cert, H4Tail, H4Improved, H4Target, G5FStar, G5Port (the former G5FStar stubs `lemmaA` and `ceslemgeneral_star` are now fully proved in G5Port; quarantine lifted), and **H4Dichotomy** (the Thm 1.3 port, de-quarantined at C4-P1: dense regime, both (K) double-counts, the per-o-pair `K_epairs_le2` + `balanced_kill` collapse, `T_star`, `triangle_free_fib/cap_fib`, `structural_case`, and `h4_eq_4` all kernel-closed; the 92-structure packing certificates retained as belt). Paper-item map: Thm 1.1 = `h4_le_1000` [K]; Thm 1.2 = `four_le_hFun_four`, `gFun_four_lt_hFun_four` [K]; **Thm 1.3 = `h4_eq_4` [K]** (corollaries `h4_eq_4_charter`, `h4_eq_4_c2`); Thm 1.4 = `g5upper_star_charter` (+ fallback `g5upper_star`, targets `T1TargetG5_closed_charter / _3520600`, engine `lemmaA` + `ceslemgeneral_star` at general k, numeric layer `key_ineq_star*`, `fStar5_le_poly`) [K]; Thm 1.5 = `fibCnt_lt_hFun_five` [K]. Upstream anchors cited: `weak_sidon_key_ineq`, `weak_sidon_bound`, `ceslemgeneral_pos`, `ceslemgeneral`, `h4upper`, `g5upper`, `h5lower`, `g5lower`, `g4`, `gFun_le_hFun_le_gFun_succ`, `generalupper`. **Audit coverage:** every theorem named in this appendix — including each upstream ingredient cited as [K] in the body (`weak_sidon_key_ineq` in Lemma 4.1, `ceslemgeneral_pos` in §6) — appears explicitly in `lean/scripts/auditC4Synth.log` (28 declarations + compile-time statement pins, incl. $331777 = 24^4 + 1$ and `hFun`-by-`rfl`; the H4Dichotomy declarations additionally in the module-scoped `auditC4P1.log`) with axioms exactly `[propext, Classical.choice, Quot.sound]` (the `decide`-backed `dichoCerts_ok` drops `Classical.choice`) and zero sorryAx, against a from-scratch rebuild (`c4synth_build.log`, 8039 jobs).

Appendix D. Reproducibility

Every numeric claim has a single re-run entry point (Appendix B, last column). SAT pipeline tooling: CaDiCaL (reference binary), z3, drat-trim, lrat-check, cake_lpr (binary built from repo-pinned source, sha256 in archive manifest). All long-running compute at OS idle priority.

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